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CLEARVIEW

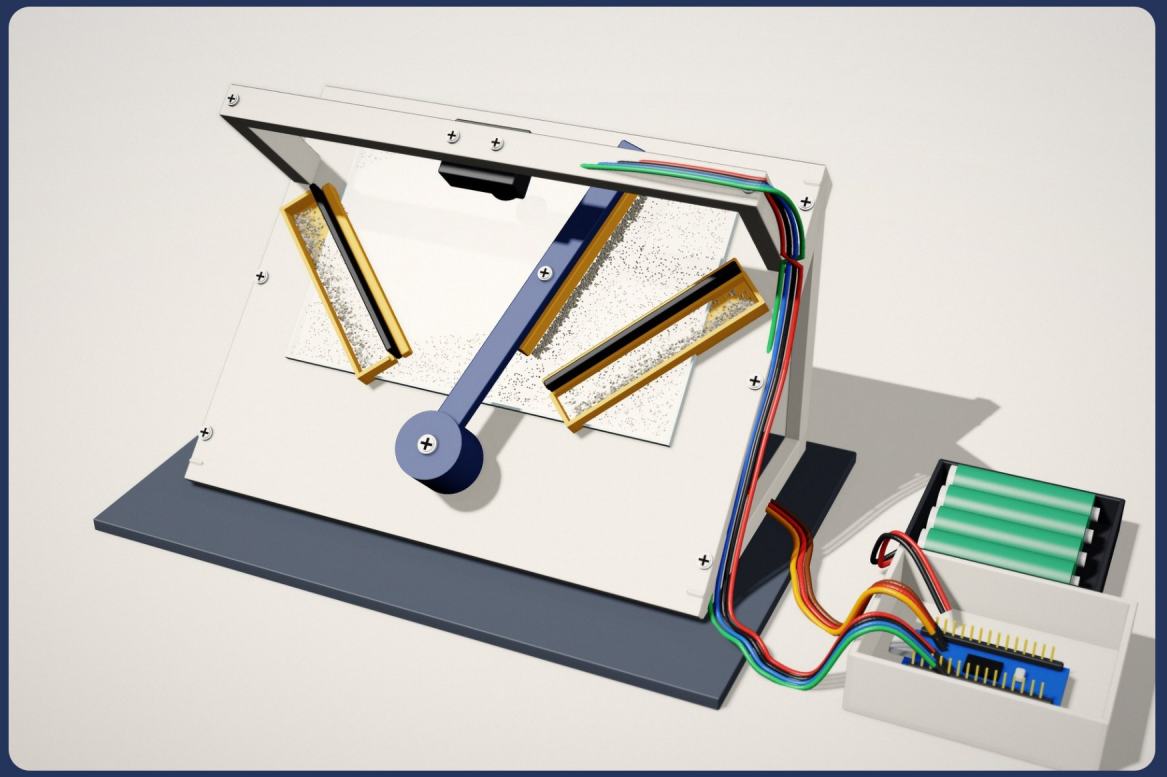
A spacesuit visor wiper with a self-cleaning edge

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Section 2 · Crewed Spaceflight · Tashkent



Visualization of the model, not a photo

APOLLO 17 · DECEMBER 1972

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I think dust is probably one of the greatest inhibitors to a nominal operation on the Moon.

Gene Cernan, commander of Apollo 17

Post-flight technical debriefing, 1973



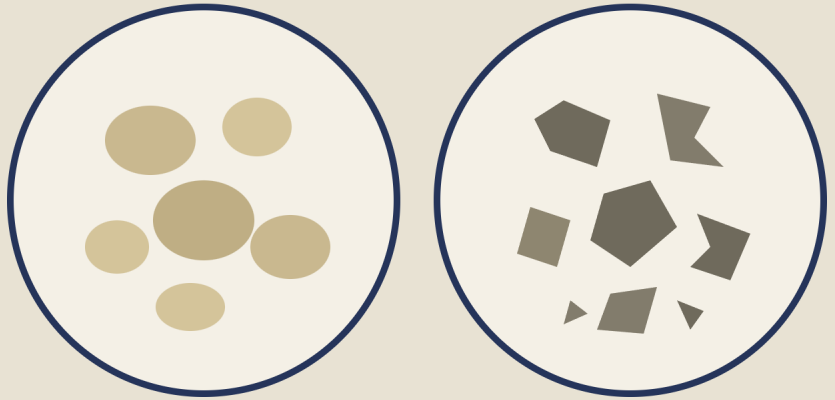
Cernan in the lunar module after a moonwalk: the suit is grey with dust. Photo: NASA

THE PROBLEM

LUNAR DUST VS. THE VISOR

FINE	finer than a human hair, it gets into every gap
SHARP	the Moon has no wind or water, so the grains never get rounded like sand
CHARGED	sunlight charges the dust, and it clings to surfaces

Apollo 17: the astronauts kept wiping dust off their visors. Artemis is sending people back to the Moon, and the dust is still there.



Earth sand: rounded **Lunar dust: sharp**

Particle shapes under magnification. The sharp edges are what scratch the visor.

EXISTING SOLUTIONS

WHAT HAS BEEN TRIED



APOLLO BRUSHES

USED ON THE MOON

+ simple, no electronics

– scratch the visor and clog with dust



NASA BRUSH STUDY

LAB TESTS, 2011

+ the best brushes removed over 90% of the dust

– tested on radiator surfaces, not visors



NASA DUST SHIELD

ELECTRODYNAMIC

+ an electric field pushes dust off without touching

– complex electronics, high voltage

THE GAP I could not find a comparison of wiper materials made for a visor, judged by transparency and scratches

GOAL AND HYPOTHESIS

WHAT I WANT TO TEST

GOAL

Find out on a model whether an edge that cleans itself keeps the visor clear over many wipes without damaging it, and pick the best of three materials.

HYPOTHESIS

An edge that gets scraped clean keeps the visor clear longer than one that does not, and soft silicone scratches less than a stiff brush.

01

Study the dust problem and what has been tried

02

Design and 3D-print a model with two scraper-troughs

03

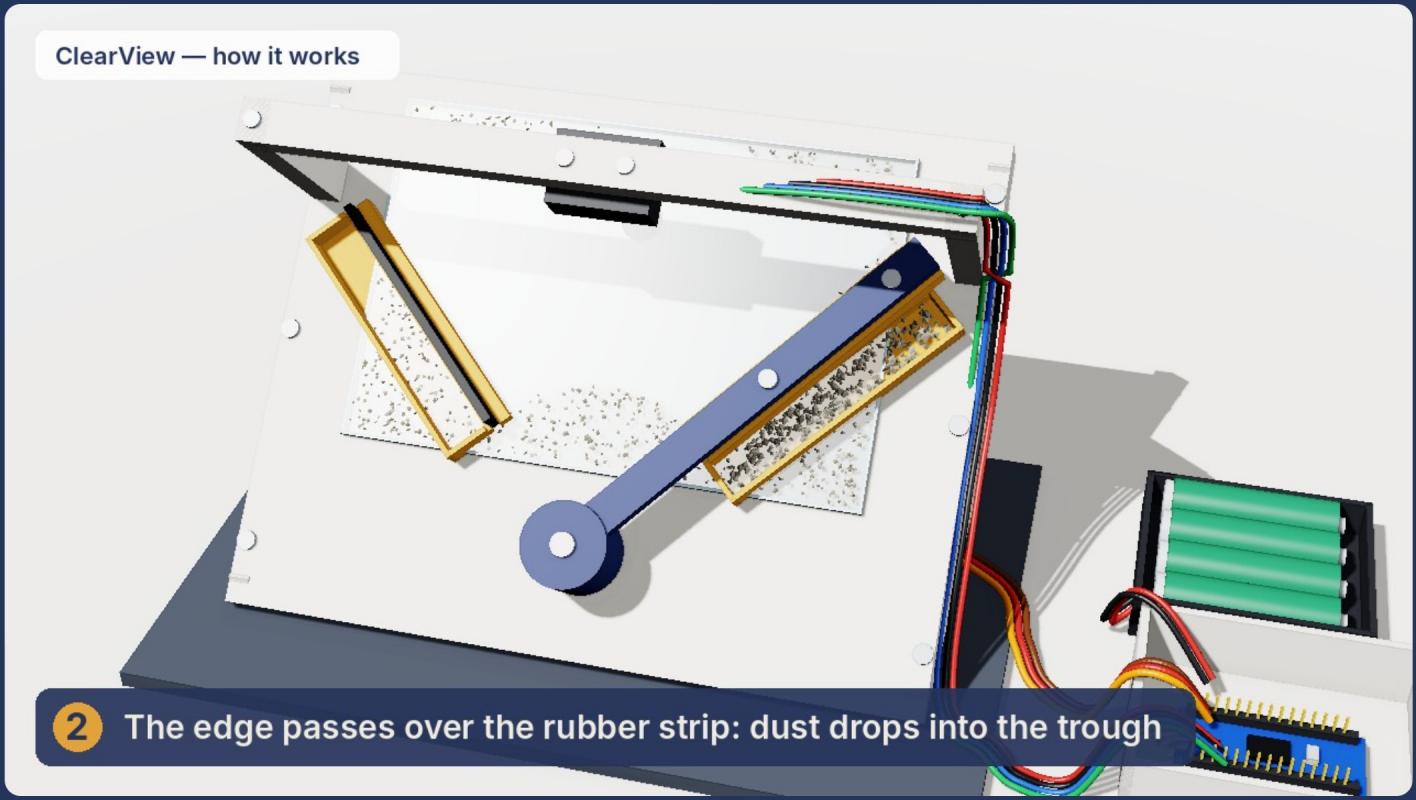
Test 6 variants, 18 runs in total

04

Compare the results and recommend a material

THE CLEARVIEW IDEA

THE EDGE CLEANS ITSELF



A plain wiper edge clogs and smears the dust back. Here the edge gets cleaned in every cycle.

1 · WIPE

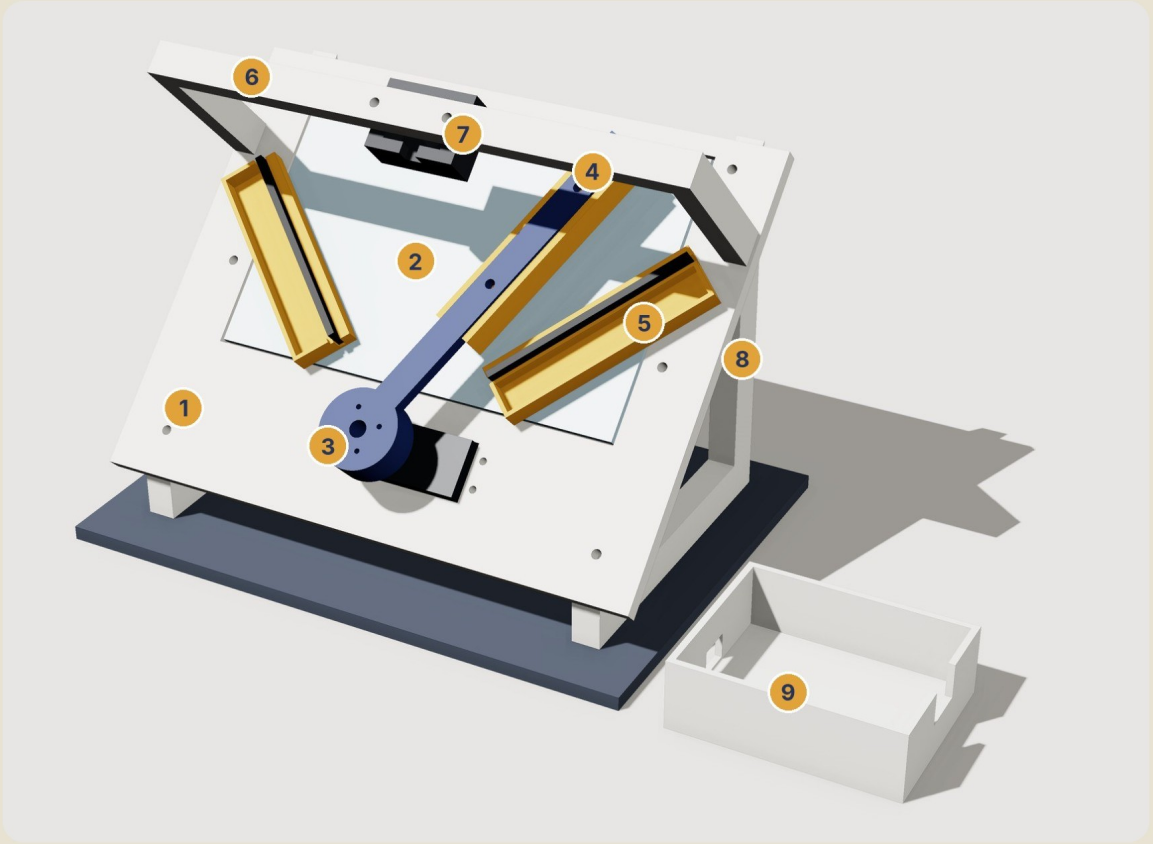
The edge pushes the dust across the plate.

2 · SCRAPE

It passes over a rubber strip, and the dust drops into the trough.

3 · BACK AGAIN

The return stroke starts clean. The rest goes into the second trough.



THE MODEL

WHAT IT IS MADE OF

1 · Tilted panel, set at 45°

2 · Acrylic plate 150×100 mm stands in for the visor

3 · MG996R servo behind the panel swings the arm ±42°

4 · Arm and holder for a brush, microfibre or silicone strip

5 · Two scraper-troughs clean the edge and catch the dust

6 · Arch over the plate

7 · BH1750 light sensor in a black housing; an LED shines from behind the panel

8 · Side supports and base

9 · Arduino Nano case

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14 parts printed in PLA. The picture is a render of the print files.

MEASUREMENT

MEASURING TRANSPARENCY

$$T = E \div E_0 \times 100\%$$

E is the light reaching the sensor now

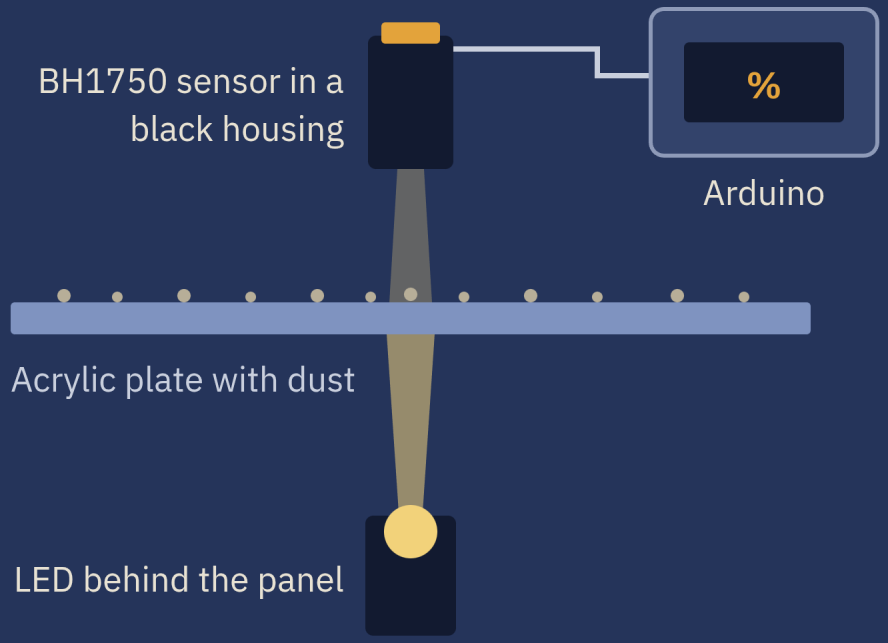
E_0 is the light with a clean plate

100%

clean plate

0%

sensor covered

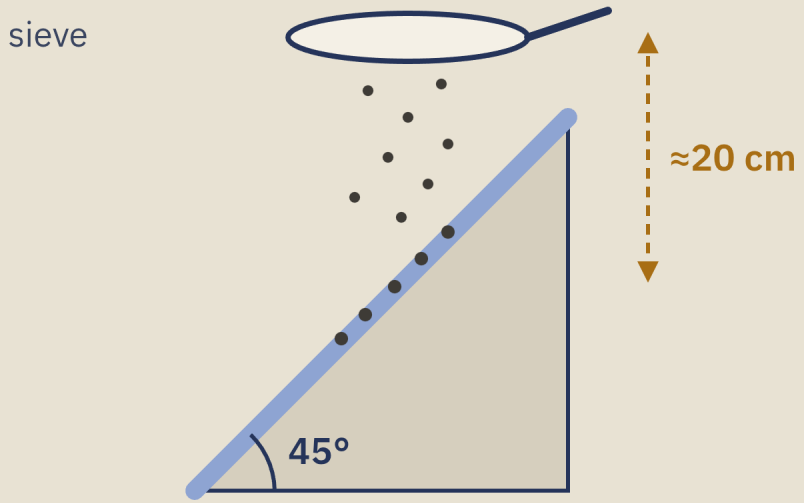


METHOD

HOW THE EXPERIMENT RUNS

Edge material	With scraper	Without scraper
Stiff brush	3 runs	3 runs
Microfibre	3 runs	3 runs
Soft silicone	3 runs	3 runs

6 variants × 3 runs = 18 runs. Results are averaged.



≈1 g of gabbro-diabase dust per run

Dust from both troughs weighed to **0.01 g**

Readings after **1, 5, 10, 20** strokes

50 strokes for the scratch test

WORK IN PROGRESS

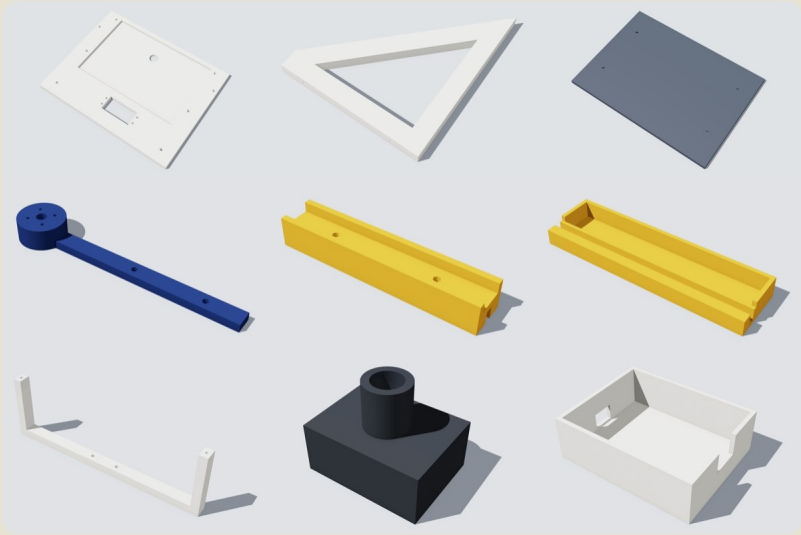
PROGRESS

DONE

- All 14 parts designed; every screw hole and the light path checked on the computer before printing
- All 3D parts printed
- Dust simulant chosen: crushed gabbro-diabase
- All electronics bought: Arduino Nano, MG996R servo, BH1750 sensor, two 18650 cells with a 6 V step-down

NEXT

- Assemble the model
- Write the Arduino code and calibrate the sensor
- Run the 18 tests and fill in the results



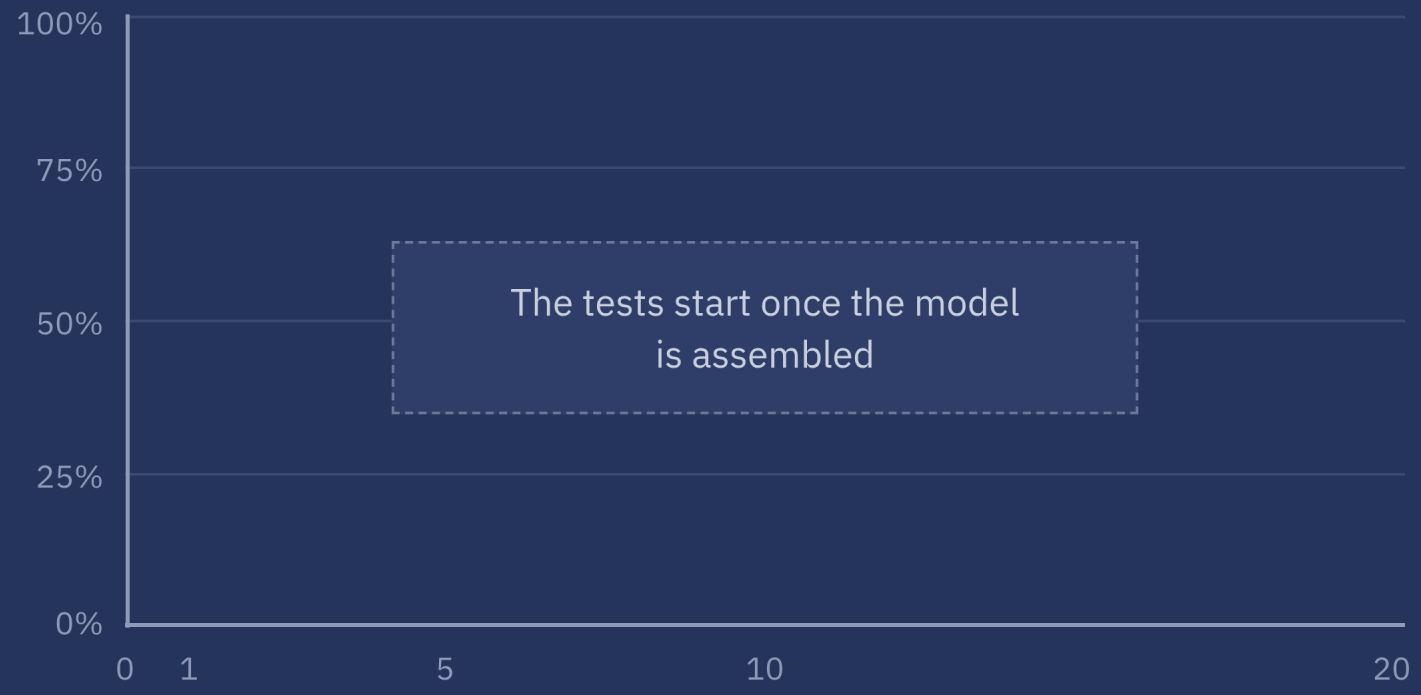
The printed parts, rendered from the print files



One of the printed parts: the material holder

WHAT I WILL MEASURE

EXPECTED RESULTS



THREE QUESTIONS

How fast does transparency drop without the scraper?

How much does the scraper slow that down?

Which material keeps the plate clear longest?

Solid line: with scraper. Dashed: without. Horizontal axis: number of strokes.

WHAT THE MODEL CAN'T SHOW

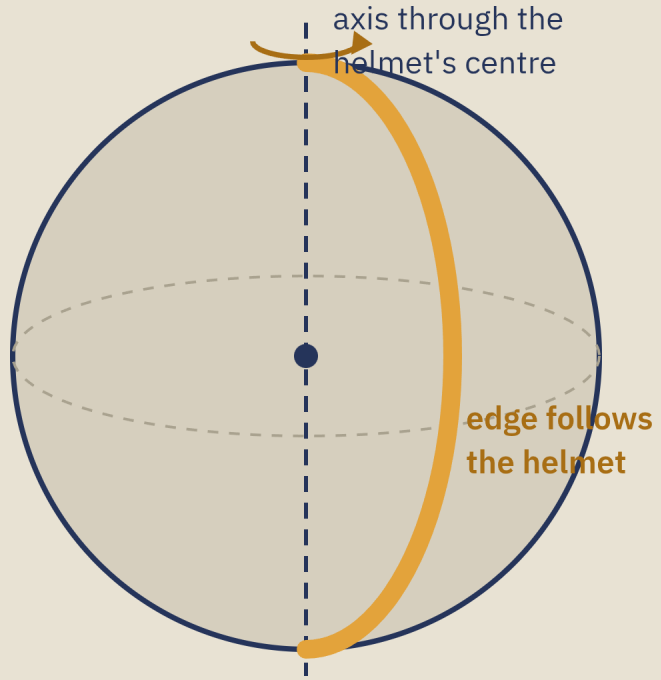
LIMITS AND NEXT STEPS

LIMITS

- Earth gravity, six times the Moon's
- Air, not vacuum
- Gabbro-diabase dust carries no electric charge
- The plate is flat

NEXT STEPS

- A curved edge for a real helmet
- A button on the glove
- Pair it with NASA's dust shield



A helmet is part of a sphere. A curved edge turning around an axis through its centre stays pressed to the visor in every position.

THANK YOU!

Happy to take your questions

SOURCES

1. NASA: Scientists Developing Ways to Mitigate Dust Problem for Explorers
2. Acta Astronautica: Flexible electrodynamic dust shields for lunar missions, 2026
3. Gaier et al., Evaluation of Brushing as a Lunar Dust Mitigation Strategy, NASA, 2011
4. Kobrick et al., Three-Body Abrasion Testing Using Lunar Dust Simulants, NASA, 2010
5. Schrader et al., Modal Analyses of Lunar Regolith Simulants incl. MLS-1, NASA JSC, 2008
6. Apollo 17 Technical Crew Debriefing, NASA, 1973; quoted in Levine (ed.), The Impact of Lunar Dust on Human Exploration

[QR code: video of the model working]

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